6. N-Gram Model for Text Generation Using Bigram

Aim: To implement a bigram model in Python to generate text based on the previous word.

Procedure:

1. Take a sample sentence.
2. Split the sentence into words.
3. Create bigram pairs of consecutive words.
4. Select a starting word.
5. Generate and display text.

Code:

```
import random
text = "i like python i like coding i love python"
words = text.split() bigrams = {}
for i in range(len(words) - 1):
    bigrams.setdefault(words[i], []).append(words[i + 1])
word = "i"
result = [word]
for i in range(5):
    if word not in bigrams:
        break
    word = random.choice(bigrams[word])
    result.append(word)
print("Generated Text:", " ".join(result))
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/6.py"
Generated Text: i like coding i like coding
```

Result: The bigram model successfully generated text using consecutive word pairs.